Module 2: FSHD Clinical Trial Project

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Project Goal:

Our goal is to help choose useful measurements for an early clinical trial of a new FSHD treatment. Our main question is: How does MRI fat fraction change over time, and how can it be used alongside strength and movement tests to track FSHD progression? Evaluating the usefulness of a measurement can help us detect changes in disease progression and determine whether a therapy is benefiting the patient.

To begin answering this question, we analyzed MRI images of the left tibialis anterior from one participant at two visits (TP1 and TP2). We calculated fat fraction at eight locations along the muscle and compared the results between visits. This helps us explore fat fraction as a measure of changes in muscle composition.

Disease Background

- **Prevalence and incidence:** Facioscapulohumeral muscular dystrophy (FSHD) is a disease characterized by muscle weakness. Signs and symptoms usually appear during adolescence, but the age of onset and severity vary widely. According to the National Library of Medicine, approximately 1 in 20,000 people have FSHD; about 95% of cases are FSHD1, and the remaining 5% are FSHD2. DelveInsight estimated that 33,500 people were living with FSHD in the United States in 2021. These are prevalence estimates. A Dutch study estimated an annual incidence of approximately 3 newly diagnosed cases per million people (Deenen et al., 2014; DelveInsight, 2025; National Library of Medicine, n.d.-a).
- **Symptoms:** FSHD often affects the face, shoulders, and upper arms, but it can also affect the legs and other muscles. People may have difficulty lifting their arms, closing their eyes, smiling, walking, or climbing stairs. Weakness may be more severe on one side than the other. Weak shoulder muscles can also cause the shoulder blades to stick out. Additional symptoms may include hearing loss and retinal abnormalities that can affect vision (National Library of Medicine, n.d.-a).

- **Diagnosis:** Doctors examine the pattern of muscle weakness and use genetic testing to help confirm FSHD. MRI can show changes within a muscle, but an MRI fat-fraction value alone does not diagnose the disease. Doctors consider symptoms, family history, and test results and may perform additional tests to rule out other conditions (Preston & Wang, 2025).

- **Current care:** Treatment focuses on managing symptoms and helping people maintain independence. Physical therapy, appropriate exercise, pain management, and braces for foot weakness can help with movement and daily activities. Some patients need breathing support, while selected patients may benefit from surgery to stabilize the shoulder blades. These approaches address the effects of the disease rather than its underlying genetic cause (Preston & Wang, 2025).

- **Progression and prognosis:** FSHD is a slowly progressing disease, but its severity and rate of progression vary among patients. For some, it may cause only mild muscle weakness, while others may lose the ability to walk due to FSHD. Generally, FSHD does not affect life expectancy, although complications involving breathing can occur in some cases. Since the disease is lifelong and progresses slowly, it is important to evaluate changes over an extended period using multiple measurements (National Library of Medicine, n.d.-b).

- **Biological mechanisms:** FSHD involves abnormal activity of the DUX4 gene in skeletal muscle. In FSHD1, a shortened region of repetitive DNA called D4Z4 allows DUX4 to become active when it would normally be silenced. This abnormal activity contributes to muscle damage and weakness, leading to muscle atrophy, inflammation, and the replacement of muscle tissue with fat and scar tissue (National Library of Medicine, n.d.-a).

- **Clinical trials and future treatments:** Current research is focused on addressing the underlying causes of FSHD. One example is delpacibart braxlosiran (del-brax) from Novartis, which belongs to a new class of RNA therapeutics and has shown potential to become the first disease-modifying treatment for FSHD. It has met its primary biomarker endpoint in Phase I/II clinical trials, and Phase III is currently underway. Del-brax uses an antibody oligonucleotide conjugate to deliver small interfering RNA to muscle cells, where it is intended to suppress DUX4 expression (Novartis, 2026).

Data Set

This project uses an FSHD dataset provided for BME 2315 that includes MRI measurements, clinical assessments, and functional measurements related to disease progression. The clinical dataset is derived from the research published by Wong et al. (2024), titled *Regional and bilateral MRI and gene signatures in facioscapulohumeral dystrophy: implications for clinical trial design and mechanisms of disease progression*, in the journal *Human Molecular Genetics*. The study investigated MRI measurements, gene expression signatures, and clinical characteristics associated with FSHD to better understand disease progression and inform clinical trial design.

The dataset provided for this project, *Module2_FSHD_Dataset.csv*, contains information from 34 participants, including demographic information and measurements collected at baseline, year 1, and year 2. The measurements were collected at different intervals for each participant, and not every measurement is available at every time point. The dataset allows us to investigate changes in disease-related measurements over time and compare their potential usefulness as exploratory endpoints in FSHD clinical trials.

For the MRI methodology analysis, we focus on subject UW_0020 and MRI images of the left tibialis anterior (TA) muscle. The provided MRI dataset contains images from two time points, TP1 and TP2, allowing muscle composition to be examined at different visits. The tibialis anterior is located in the anterior compartment of the lower leg and contributes to ankle dorsiflexion.

The MRI data include separate images representing water and fat signals, stored as NumPy (.npy) files. These images contain MRI signal intensities associated with water and fat within the imaged tissue. The signals are used to calculate fat fraction, which estimates the proportion of the combined water and fat signal attributable to fat. Fat fraction is expressed as a percentage, while the original image intensities represent relative signal values rather than absolute physical concentrations.

The dataset also includes segmentation masks identifying the left tibialis anterior muscle. Each mask distinguishes pixels belonging to the muscle from surrounding tissues. These masks allow fat fraction to be calculated specifically within the tibialis anterior rather than across the entire MRI image.

MRI images were provided at eight sampled locations along the muscle for each time point. The locations are expressed as percentages of muscle length, allowing measurements from different positions to be compared. The accompanying slice information file identifies the water, fat, and mask files associated with each location.

The clinical dataset includes MRI fat fraction, fat fraction variability, and STIR measurements, as well as DUX4 gene expression signatures, D4Z4 repeat information,

quantitative muscle strength measurements, and functional assessments. Functional measurements include Timed Up and Go, walking and running performance, and stair ascent and descent. Clinical severity scores are also provided.

The MRI images and segmentation masks were supplied as part of the BME 2315 course materials. The files were provided in a format that allows the water and fat images to be loaded directly into Python and the segmentation masks to be applied during analysis. The original image acquisition and segmentation procedures were not performed as part of this project. Instead, we use the supplied images and masks to reproduce the calculation of tibialis anterior fat fraction.

Data Sources

The primary research source for the clinical dataset is Wong et al. (2024), *Regional and bilateral MRI and gene signatures in facioscapulohumeral dystrophy: implications for clinical trial design and mechanisms of disease progression*, published in *Human Molecular Genetics*, Volume 33, Issue 8, pages 698–708.

Original research publication: https://doi.org/10.1093/hmg/ddae007

The data and computational materials used in this project were provided through BME 2315 and include *Module2_FSHD_Dataset.csv*, *FSHD_TA_MRI_data_0020_TP1_TP2*, and *FSHD_TA_MRI_example_notebook.ipynb*. The course-provided clinical dataset is based on the published research, while the MRI files and example notebook were supplied for the computational analysis.

MRI Data Analysis Methodology

MRI Analysis Overview

The purpose of this analysis is to calculate the fat fraction of the tibialis anterior muscle using MRI water and fat images. Fat fraction provides a measurement of muscle composition by estimating the proportion of fat signal relative to the combined water and fat signal. This measurement is relevant to FSHD because progressive muscle deterioration can result in the replacement of muscle tissue with fat.

We will use Python to load the MRI images and segmentation masks provided in the dataset. The segmentation masks will allow us to isolate the tibialis anterior muscle from surrounding tissues. We will calculate fat fraction for the entire supplied image region and for the tibialis anterior muscle specifically. We will then visualize the images,

examine the distribution of fat fraction values, and calculate summary statistics.

Finally, we will compare fat fraction measurements across different locations along the muscle and between the two available MRI time points, TP1 and TP2. This will allow us to investigate how muscle composition varies by location and visit.

Loading the MRI Dataset

We begin by importing the Python libraries needed for our analysis. NumPy will be used to process the MRI images, Pandas will organize the information associated with each MRI slice, and Matplotlib will be used to visualize the results. The slice information file identifies the subject, time point, sampling location, and corresponding water, fat, and segmentation mask files.

In [1]:
```
from pathlib import Path
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Locate the MRI data folder
DATA_DIR = Path("FSHD_MRI_data/FSHD_TA_MRI_data_0020_TP1_TP2")

# Load the MRI slice information
manifest = pd.read_csv(DATA_DIR / "slice_info.csv")

# Display the dataset information
display(manifest)
```

	subject	timepoint	slice_id	percent_along_muscle	z_index	distance_from_muscl
0	UW_0020	TP1	1	10	41	
1	UW_0020	TP1	2	20	46	
2	UW_0020	TP1	3	30	52	
3	UW_0020	TP1	4	40	58	
4	UW_0020	TP1	5	50	64	
5	UW_0020	TP1	6	60	69	
6	UW_0020	TP1	7	70	75	
7	UW_0020	TP1	8	80	81	
8	UW_0020	TP2	1	10	40	
9	UW_0020	TP2	2	20	46	
10	UW_0020	TP2	3	30	52	
11	UW_0020	TP2	4	40	58	
12	UW_0020	TP2	5	50	64	
13	UW_0020	TP2	6	60	71	
14	UW_0020	TP2	7	70	77	
15	UW_0020	TP2	8	80	83	

Loading MRI Images and Segmentation Masks

We begin by examining the first MRI slice from TP1. Each slice contains a water image, a fat image, and a segmentation mask identifying the tibialis anterior muscle. The water and fat images contain MRI signal intensities, while the segmentation mask identifies which pixels belong to the muscle. By applying the mask, we can calculate fat fraction specifically within the tibialis anterior rather than including surrounding tissues.

In [2]:
```
# Select the first MRI slice from TP1
slice_data = manifest[manifest["timepoint"] == "TP1"].iloc[0]

# Load the water, fat, and segmentation mask images
water = np.load(DATA_DIR / slice_data["water_file"])
fat = np.load(DATA_DIR / slice_data["fat_file"])
mask = np.load(DATA_DIR / slice_data["mask_file"])

# Convert the segmentation mask to Boolean values
```

```
muscle_mask = mask > 0

# Display image information
print("Water image shape:", water.shape)
print("Fat image shape:", fat.shape)
print("Mask shape:", mask.shape)
print("Number of muscle pixels:", np.sum(muscle_mask))
```

```
Water image shape: (266, 224)
Fat image shape: (266, 224)
Mask shape: (266, 224)
Number of muscle pixels: 61
```

Calculating Fat Fraction

Fat fraction is calculated by dividing the fat signal intensity by the combined fat and water signal intensities. This calculation produces a percentage representing the proportion of the total signal associated with fat. To avoid division by zero, pixels where the combined water and fat signal is zero are excluded from the calculation. The segmentation mask is then used to isolate fat fraction values within the tibialis anterior muscle.

In [3]:
```
# Calculate the combined water and fat signal
total_signal = water + fat

# Calculate fat fraction while avoiding division by zero
fat_fraction = np.divide(
    fat,
    total_signal,
    out=np.zeros_like(fat, dtype=float),
    where=total_signal > 0
) * 100

# Extract fat fraction values within the tibialis anterior
muscle_fat_fraction = fat_fraction[muscle_mask]

# Calculate summary statistics
mean_ff = np.mean(muscle_fat_fraction)
median_ff = np.median(muscle_fat_fraction)
std_ff = np.std(muscle_fat_fraction)

print(f"Mean fat fraction: {mean_ff:.2f}%")
print(f"Median fat fraction: {median_ff:.2f}%")
print(f"Standard deviation: {std_ff:.2f}%")
print(f"Number of muscle pixels: {len(muscle_fat_fraction)}")
```

```
Mean fat fraction: 83.92%
Median fat fraction: 88.83%
Standard deviation: 13.27%
Number of muscle pixels: 61
```

MRI Visualization

To visualize the MRI data, we display the water image, fat image, segmentation mask, and calculated fat fraction map. The water and fat images show the distribution of their respective MRI signals, while the segmentation mask identifies the tibialis anterior muscle. The fat fraction map displays the estimated percentage of fat signal at each pixel. Comparing these images helps demonstrate how the original MRI data are used to calculate muscle fat fraction.

In [4]:
```
# Create a figure with four MRI images
fig, axes = plt.subplots(1, 4, figsize=(16, 4))

# Water image
axes[0].imshow(water, cmap="gray")
axes[0].set_title("Water Image")

# Fat image
axes[1].imshow(fat, cmap="gray")
axes[1].set_title("Fat Image")

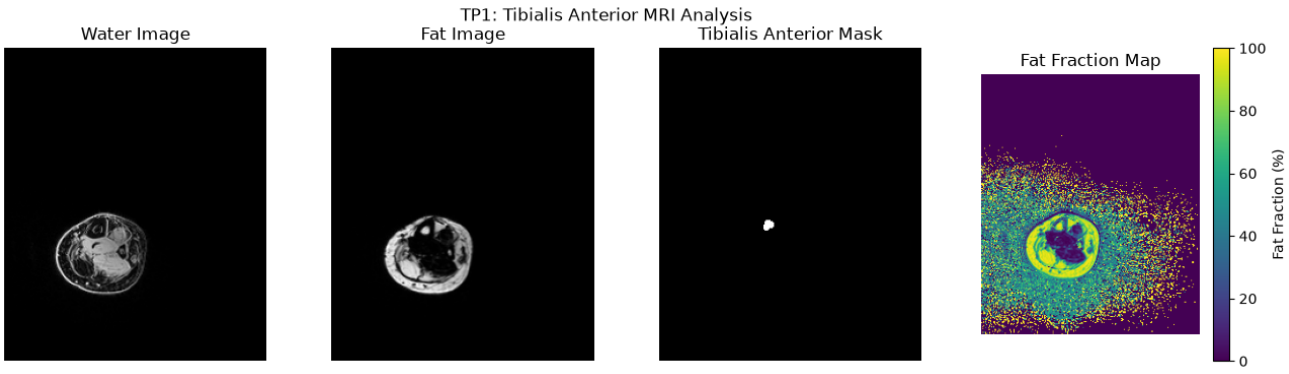
# Segmentation mask
axes[2].imshow(muscle_mask, cmap="gray")
axes[2].set_title("Tibialis Anterior Mask")

# Fat fraction map
im = axes[3].imshow(fat_fraction, cmap="viridis", vmin=0, vmax=100)
axes[3].set_title("Fat Fraction Map")

# Remove axes for clearer MRI visualization
for ax in axes:
    ax.axis("off")

# Add a color bar showing fat fraction percentage
fig.colorbar(im, ax=axes[3], label="Fat Fraction (%)")

plt.suptitle("TP1: Tibialis Anterior MRI Analysis")
plt.show()
```



Interpretation of Initial MRI Results

The first MRI slice from TP1 produced a mean fat fraction of 83.92% and a median fat fraction of 88.83% within the segmented tibialis anterior muscle. The standard deviation was 13.27 percentage points, indicating variability in fat fraction across the muscle pixels. The segmentation mask contained 61 pixels, allowing the analysis to focus specifically on the tibialis anterior rather than surrounding tissues.

The relatively high fat fraction suggests that the segmented region contains a substantial fat signal compared with its water signal. However, these results represent only one slice from one time point. Additional slices and the second time point must be analyzed to better understand how fat fraction varies along the muscle and between visits.

Distribution of Muscle Fat Fraction

To further examine the variation in fat fraction, we created a histogram of the fat fraction values within the segmented tibialis anterior muscle. This allows us to visualize the frequency of different fat fraction percentages and determine whether the values are concentrated around the mean or distributed across a wider range.

In [5]:
```
# Plot the distribution of fat fraction within the muscle
plt.figure(figsize=(8, 5))

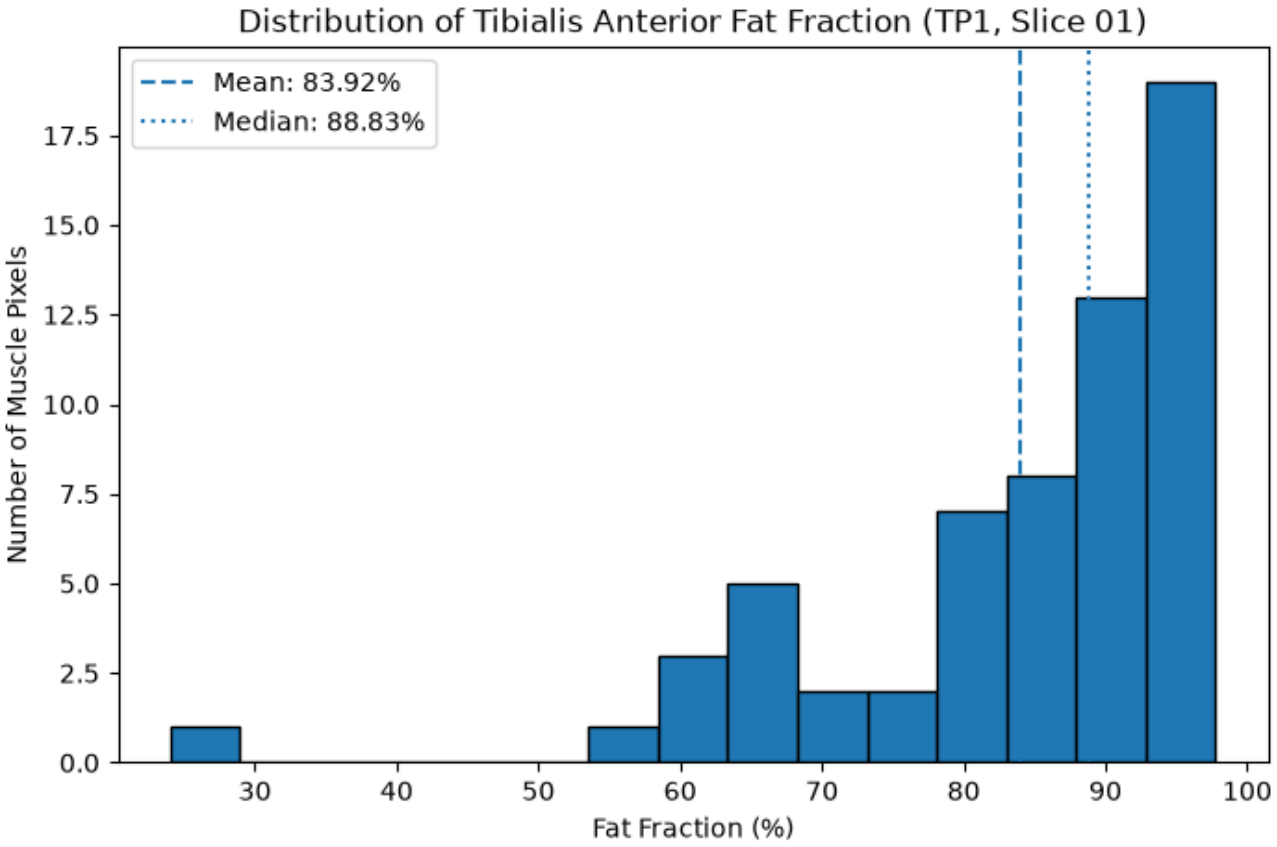
plt.hist(muscle_fat_fraction, bins=15, edgecolor="black")

# Add lines for the mean and median
plt.axvline(mean_ff, linestyle="--", label=f"Mean: {mean_ff:.2f}%")
plt.axvline(median_ff, linestyle=":", label=f"Median: {median_ff:.2f}%")

plt.xlabel("Fat Fraction (%)")
plt.ylabel("Number of Muscle Pixels")
plt.title("Distribution of Tibialis Anterior Fat Fraction (TP1, Slice 01)")
```

```
plt.legend()

plt.show()
```



Interpretation of Fat Fraction Distribution

The histogram shows that most pixels within the segmented tibialis anterior muscle have fat fraction values between approximately 80% and 100%. The mean fat fraction was 83.92%, while the median was 88.83%. The distribution is left-skewed, with several pixels having lower fat fractions that decrease the mean. This suggests that the fat signal is relatively high throughout much of the segmented region, although there is some variation between pixels. These results describe only the first slice at TP1 and cannot establish whether fat fraction changes over time.

Fat Fraction Across MRI Slices and Time Points

To investigate how fat fraction varies along the tibialis anterior muscle, we calculated the mean fat fraction for each of the eight MRI slices at TP1 and TP2. Each slice represents a different location along the muscle. The segmentation masks were used to isolate the tibialis anterior, and the mean fat fraction was calculated using only pixels within the

segmented region. The resulting measurements allow us to compare muscle composition across locations and between the two MRI visits.

In [6]:
```
# Store the results for each MRI slice
results = []

# Analyze every slice in the dataset
for _, row in manifest.iterrows():

    # Load the MRI images and segmentation mask
    water = np.load(DATA_DIR / row["water_file"])
    fat = np.load(DATA_DIR / row["fat_file"])
    mask = np.load(DATA_DIR / row["mask_file"])

    # Identify pixels belonging to the tibialis anterior
    muscle_mask = mask > 0

    # Calculate fat fraction
    total_signal = water + fat

    fat_fraction = np.divide(
        fat,
        total_signal,
        out=np.zeros_like(fat, dtype=float),
        where=total_signal > 0
    ) * 100

    # Extract fat fraction values within the muscle
    muscle_values = fat_fraction[muscle_mask]

    # Calculate summary statistics
    results.append({
        "Time Point": row["timepoint"],
        "Muscle Location (%)": row["percent_along_muscle"],
        "Mean Fat Fraction (%)": np.mean(muscle_values),
        "Median Fat Fraction (%)": np.median(muscle_values),
        "Standard Deviation": np.std(muscle_values),
        "Muscle Pixels": len(muscle_values)
    })

# Convert results into a DataFrame
results_df = pd.DataFrame(results)

# Display the results
display(results_df.round(2))
```

	Time Point	Muscle Location (%)	Mean Fat Fraction (%)	Median Fat Fraction (%)	Standard Deviation	Muscle Pixels
0	TP1	10	83.92	88.83	13.27	61
1	TP1	20	82.71	86.65	12.43	115
2	TP1	30	58.50	61.82	22.88	231
3	TP1	40	29.93	25.37	21.58	360
4	TP1	50	20.87	14.55	20.06	421
5	TP1	60	16.76	9.05	19.67	440
6	TP1	70	17.51	8.96	20.46	388
7	TP1	80	27.87	22.69	22.63	263
8	TP2	10	82.39	90.84	15.91	46
9	TP2	20	84.05	92.03	17.74	74
10	TP2	30	81.82	86.65	14.73	170
11	TP2	40	55.45	56.74	22.15	242
12	TP2	50	37.15	36.05	23.39	321
13	TP2	60	28.35	24.15	20.39	367
14	TP2	70	25.95	21.56	19.10	320
15	TP2	80	40.14	37.00	18.94	217

In [7]:
```
# Verify segmentation masks for all 16 MRI slices
for _, row in manifest.iterrows():
    mask_check = np.load(DATA_DIR / row["mask_file"])
    labels = np.unique(mask_check)
    print(row["timepoint"], row["percent_along_muscle"], labels)
```

```
TP1 10 [0 1]
TP1 20 [0 1]
TP1 30 [0 1]
TP1 40 [0 1]
TP1 50 [0 1]
TP1 60 [0 1]
TP1 70 [0 1]
TP1 80 [0 1]
TP2 10 [0 1]
TP2 20 [0 1]
TP2 30 [0 1]
TP2 40 [0 1]
TP2 50 [0 1]
TP2 60 [0 1]
TP2 70 [0 1]
TP2 80 [0 1]
```

Comparison of Fat Fraction Across Muscle Locations

To compare muscle composition between the two MRI visits, we plotted the mean fat fraction at each sampled location along the tibialis anterior muscle. This allows us to examine whether fat fraction varies along the muscle and whether differences are observed between TP1 and TP2.

In [8]:
```
# Separate results by time point
tp1 = results_df[results_df["Time Point"] == "TP1"]
tp2 = results_df[results_df["Time Point"] == "TP2"]

# Sort measurements by muscle location
tp1 = tp1.sort_values("Muscle Location (%)")
tp2 = tp2.sort_values("Muscle Location (%)")

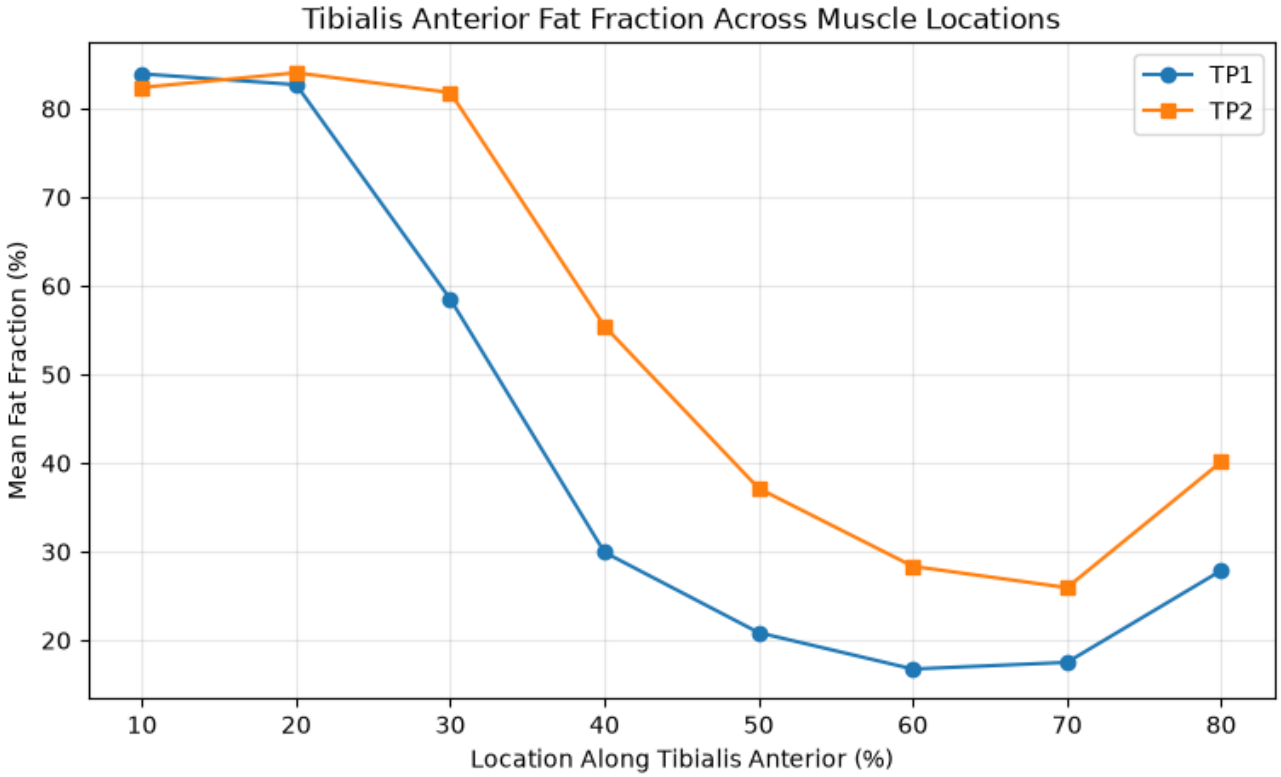
# Plot mean fat fraction for both visits
plt.figure(figsize=(9, 5))

plt.plot(
    tp1["Muscle Location (%)"],
    tp1["Mean Fat Fraction (%)"],
    marker="o",
    label="TP1"
)

plt.plot(
    tp2["Muscle Location (%)"],
    tp2["Mean Fat Fraction (%)"],
    marker="s",
    label="TP2"
)
```

```
plt.xlabel("Location Along Tibialis Anterior (%)")
plt.ylabel("Mean Fat Fraction (%)")
plt.title("Tibialis Anterior Fat Fraction Across Muscle Locations")
plt.legend()
plt.grid(alpha=0.3)

plt.show()
```



Interpretation of Fat Fraction Across Muscle Locations

The graph shows that mean fat fraction varies considerably along the tibialis anterior muscle. At both time points, fat fraction was highest near the beginning of the sampled muscle length and generally decreased toward 60% to 70% of muscle length. Fat fraction increased again at the 80% location.

TP2 generally showed higher mean fat fraction values than TP1, particularly between 30% and 60% of muscle length. For example, at 30% of muscle length, mean fat fraction increased from approximately 58% at TP1 to 82% at TP2. This suggests a difference in muscle composition between the two visits.

However, these measurements represent a single participant, and differences between visits may also be influenced by image acquisition, segmentation, or sampling variability. Therefore, additional measurements and participants would be needed to determine

whether these changes reliably reflect FSHD progression.

Overall Comparison Between MRI Visits

To summarize the differences between MRI visits, we calculated the average of the eight mean fat fraction measurements for TP1 and TP2. We then calculated the difference between the two time points. This provides a summary of the average fat fraction across the sampled muscle locations, allowing us to quantify the overall difference between visits.

In [9]:
```
# Calculate the average of the eight slice means
tp1_mean = tp1["Mean Fat Fraction (%)"].mean()
tp2_mean = tp2["Mean Fat Fraction (%)"].mean()

# Calculate the change between visits
difference = tp2_mean - tp1_mean

# Display results
print(f"TP1 average fat fraction: {tp1_mean:.2f}%")
print(f"TP2 average fat fraction: {tp2_mean:.2f}%")
print(f"Difference between visits: {difference:+.2f} percentage points")
```

```
TP1 average fat fraction: 42.26%
TP2 average fat fraction: 54.41%
Difference between visits: +12.15 percentage points
```

Interpretation of Overall MRI Results

The average fat fraction across the eight sampled muscle locations increased from 42.26% at TP1 to 54.41% at TP2, representing an increase of 12.15 percentage points. This suggests that the tibialis anterior muscle had a greater proportion of fat signal at the second MRI visit.

The results also showed that fat fraction varied considerably along the length of the muscle, indicating that muscle composition was not uniform across the sampled locations. These findings suggest that MRI fat fraction may be a useful measurement for evaluating changes in muscle composition in patients with FSHD.

However, this analysis only examined one participant, and the averages were calculated by giving equal weight to each of the eight sampled slices. Therefore, these results should not be interpreted as representing the entire muscle volume or the broader FSHD population. Additional participants and measurements would be necessary to determine whether the observed increase consistently reflects disease progression.

Interpolation Analysis:

Interpolation of Tibialis Anterior Fat Fraction

The MRI dataset contains fat fraction measurements at eight locations along the tibialis anterior muscle, ranging from 10% to 80% of muscle length in increments of 10%. However, fat fraction values are not directly measured at locations between these sampled points.

To estimate fat fraction at an unsampled location, we will use linear interpolation. This method estimates a value between two known measurements by assuming that fat fraction changes at a constant rate between those locations.

For this analysis, we will estimate the mean fat fraction at 45% of muscle length using the measured values at 40% and 50%. We will perform this calculation for both TP1 and TP2 to compare the estimated fat fraction between visits.

Because 45% falls between two measured locations, this calculation is interpolation rather than extrapolation.

In [10]:
```
# Define the location where fat fraction will be estimated
target_location = 45

# Interpolate fat fraction at 45% muscle length for TP1
tp1_interpolated = np.interp(
    target_location,
    tp1["Muscle Location (%)"],
    tp1["Mean Fat Fraction (%)"]
)

# Interpolate fat fraction at 45% muscle length for TP2
tp2_interpolated = np.interp(
    target_location,
    tp2["Muscle Location (%)"],
    tp2["Mean Fat Fraction (%)"]
)

# Display the estimated fat fractions
print(f"TP1 estimated fat fraction at 45%: {tp1_interpolated:.2f}%")
print(f"TP2 estimated fat fraction at 45%: {tp2_interpolated:.2f}%")
```

```
TP1 estimated fat fraction at 45%: 25.40%
TP2 estimated fat fraction at 45%: 46.30%
```

Visualization of Interpolated Fat Fraction

To visualize the interpolation results, we plotted the measured mean fat fraction values across the eight sampled locations for both MRI visits. We then added the interpolated estimates at 45% of muscle length.

The measured values are connected using straight lines to represent linear interpolation between sampled locations. The interpolated estimates are highlighted separately to distinguish calculated estimates from actual MRI measurements.

In [11]:
```
# Create a figure for the interpolation analysis
plt.figure(figsize=(9, 5))

# Plot measured fat fraction values for TP1
plt.plot(
    tp1["Muscle Location (%)"],
    tp1["Mean Fat Fraction (%)"],
    marker="o",
    label="TP1 Measured"
)

# Plot measured fat fraction values for TP2
plt.plot(
    tp2["Muscle Location (%)"],
    tp2["Mean Fat Fraction (%)"],
    marker="s",
    label="TP2 Measured"
)

# Highlight the interpolated estimates at 45%
plt.scatter(
    target_location,
    tp1_interpolated,
    marker="X",
    s=150,
    label=f"TP1 Estimated: {tp1_interpolated:.2f}%"
)

plt.scatter(
    target_location,
    tp2_interpolated,
    marker="X",
    s=150,
    label=f"TP2 Estimated: {tp2_interpolated:.2f}%"
)

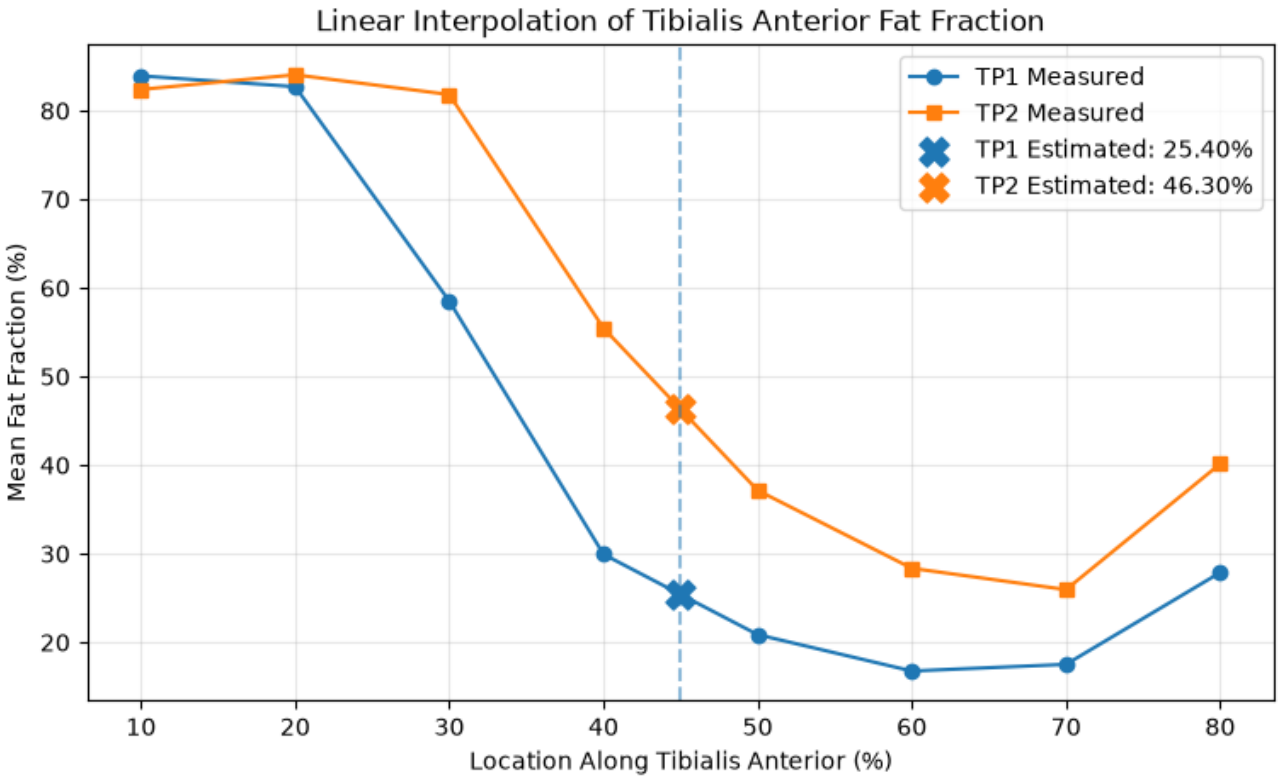
# Add a vertical reference line at the estimated location
plt.axvline(
```

```
    target_location,
    linestyle="--",
    alpha=0.5
)

# Label the graph
plt.xlabel("Location Along Tibialis Anterior (%)")
plt.ylabel("Mean Fat Fraction (%)")
plt.title("Linear Interpolation of Tibialis Anterior Fat Fraction")

plt.legend()
plt.grid(alpha=0.3)

plt.show()
```



Interpretation of Interpolation Results

Linear interpolation was used to estimate the mean fat fraction at 45% of the tibialis anterior muscle length. The estimated fat fraction was 25.40% at TP1 and 46.30% at TP2, representing a difference of 20.90 percentage points between the two visits.

The estimates were calculated using the measured mean fat fraction values at 40% and 50% of muscle length. Because 45% is halfway between these sampled locations, linear interpolation estimates the fat fraction by taking the average of the two neighboring measurements.

The interpolation method assumes that fat fraction changes at a constant rate between the sampled locations. However, the actual distribution of fat within the muscle may not follow a straight line. Therefore, the interpolated values are estimates rather than new MRI measurements.

Both estimates fall within the range of the sampled locations, meaning interpolation was used rather than extrapolation. Extrapolation would involve estimating values outside the measured range, where predictions would generally be more uncertain.

Several factors could affect the accuracy of these estimates, including differences in MRI acquisition, segmentation accuracy, and variation in muscle composition between sampled locations. Because no MRI measurement was collected at exactly 45% of muscle length, we cannot determine the accuracy of the interpolated estimates without additional measurements.

The higher estimated fat fraction at TP2 is consistent with the overall increase observed between visits. However, additional measurements would be needed to determine whether this difference reliably reflects FSHD progression.

Longitudinal Interpolation of Clinical Measurements

In addition to interpolating MRI fat fraction across muscle locations, we performed longitudinal interpolation using the clinical dataset.

Because participants attended follow-up visits at different times, we used linear interpolation to estimate measurements at common comparison times of 11.15 months and 22.13 months.

We then calculated mean relative percentage changes and standard deviations to compare MRI fat fraction, muscle strength, and functional assessments as potential clinical trial endpoints.

In [12]:
```
import pandas as pd

# Load the clinical dataset containing all 34 participants
clinical_df = pd.read_csv("Module2_FSHD_Dataset.csv")

# Check the dimensions and available measurements
print("Dataset shape:", clinical_df.shape)

print("\nColumn names:")
for column in clinical_df.columns:
    print(column)
```

```
display(clinical_df.head())
```

```
Dataset shape: (34, 46)

Column names:
subject_id
sex
age_baseline
d4z4
dux4_signature_baseline_R
dux4_signature_baseline_L
lag_months_baseline
lag_months_y1
lag_months_y2
ff_baseline_R
ff_y1_R
ff_y2_R
ff_baseline_L
ff_y1_L
ff_y2_L
fvar_baseline_R
fvar_y1_R
fvar_y2_R
fvar_baseline_L
fvar_y1_L
fvar_y2_L
stir_quant_baseline_R
stir_quant_y1_R
stir_quant_baseline_L
stir_quant_y1_L
dorsiflexion_qmt_R_baseline
dorsiflexion_qmt_R_y1
dorsiflexion_qmt_R_y2
dorsiflexion_qmt_L_baseline
dorsiflexion_qmt_L_y1
dorsiflexion_qmt_L_y2
tug_time_baseline
tug_time_y1
tug_time_y2
wr_10_baseline
wr_10_y1
wr_10_y2
ascend_4_baseline
ascend_4_y1
ascend_4_y2
descend_4_baseline
descend_4_y1
descend_4_y2
clinical_severity_score_baseline
clinical_severity_score_y1
clinical_severity_score_y2
```

	subject_id	sex	age_baseline	d4z4	dux4_signature_baseline_R	dux4_signature_ba
0	1-0019	M	47.06	4	0.060426	0
1	1-0020	F	24.28	4	0.029324	0
2	1-0021	M	36.42	6	0.056946	0
3	1-0022	F	59.43	2	0.877067	0
4	1-0023	M	63.92	5	2.694204	2

5 rows × 46 columns

In [13]:
```
# Display all columns with their data types and missing-value counts
column_summary = pd.DataFrame({
    "Column": clinical_df.columns,
    "Data Type": clinical_df.dtypes.values,
    "Missing Values": clinical_df.isna().sum().values
})

display(column_summary)
```

	Column	Data Type	Missing Values
0	subject_id	str	0
1	sex	str	0
2	age_baseline	float64	0
3	d4z4	int64	0
4	dux4_signature_baseline_R	float64	2
5	dux4_signature_baseline_L	float64	2
6	lag_months_baseline	int64	0
7	lag_months_y1	float64	2
8	lag_months_y2	float64	7
9	ff_baseline_R	float64	0
10	ff_y1_R	float64	4
11	ff_y2_R	float64	9
12	ff_baseline_L	float64	0
13	ff_y1_L	float64	4
14	ff_y2_L	float64	9

15	fvar_baseline_R	float64	0
16	fvar_y1_R	float64	4
17	fvar_y2_R	float64	9
18	fvar_baseline_L	float64	0
19	fvar_y1_L	float64	4
20	fvar_y2_L	float64	9
21	stir_quant_baseline_R	float64	0
22	stir_quant_y1_R	float64	14
23	stir_quant_baseline_L	float64	3
24	stir_quant_y1_L	float64	14
25	dorsiflexion_qmt_R_baseline	float64	5
26	dorsiflexion_qmt_R_y1	float64	7
27	dorsiflexion_qmt_R_y2	float64	9
28	dorsiflexion_qmt_L_baseline	float64	5
29	dorsiflexion_qmt_L_y1	float64	8
30	dorsiflexion_qmt_L_y2	float64	9
31	tug_time_baseline	float64	0
32	tug_time_y1	float64	2
33	tug_time_y2	float64	7
34	wr_10_baseline	float64	0
35	wr_10_y1	float64	2
36	wr_10_y2	float64	7
37	ascend_4_baseline	float64	0
38	ascend_4_y1	float64	2
39	ascend_4_y2	float64	7
40	descend_4_baseline	float64	0
41	descend_4_y1	float64	2
42	descend_4_y2	float64	7
43	clinical_severity_score_baseline	int64	0

44	clinical_severity_score_y1	float64	2
45	clinical_severity_score_y2	float64	7

In [14]:
```
import numpy as np
from scipy.interpolate import interp1d

# Establish common comparison times
T0 = 0
T1 = clinical_df["lag_months_y1"].min()
T2 = clinical_df["lag_months_y2"].min()

print(f"Baseline (T0): {T0:.2f} months")
print(f"Year 1 (T1): {T1:.2f} months")
print(f"Year 2 (T2): {T2:.2f} months")

# Count participants with recorded visits
print(f"\nTotal participants: {len(clinical_df)}")
print(f"Participants with Year 1 visits: {clinical_df['lag_months_y1'].notna
print(f"Participants with Year 2 visits: {clinical_df['lag_months_y2'].notna
```

```
Baseline (T0): 0.00 months
Year 1 (T1): 11.15 months
Year 2 (T2): 22.13 months

Total participants: 34
Participants with Year 1 visits: 32
Participants with Year 2 visits: 27
```

In [15]:
```
# Longitudinal interpolation of right-side MRI fat fraction

interpolated_results = []

for _, row in clinical_df.iterrows():

    # Actual measurement times
    times = np.array([
        row["lag_months_baseline"],
        row["lag_months_y1"],
        row["lag_months_y2"]
    ], dtype=float)

    # MRI fat fraction measurements
    values = np.array([
        row["ff_baseline_R"],
        row["ff_y1_R"],
        row["ff_y2_R"]
    ], dtype=float)

    # Remove missing measurements
```

```
    valid = np.isfinite(times) & np.isfinite(values)

    times = times[valid]
    values = values[valid]

    # At least two measurements are required
    if len(times) < 2:
        continue

    # Sort measurements chronologically
    order = np.argsort(times)
    times = times[order]
    values = values[order]

    # Linear interpolation
    interpolation_function = interp1d(
        times,
        values,
        kind="linear",
        bounds_error=False,
        fill_value=np.nan
    )

    # Estimate measurements at common time points
    ff_T0 = float(interpolation_function(T0))
    ff_T1 = float(interpolation_function(T1))
    ff_T2 = float(interpolation_function(T2))

    interpolated_results.append({
        "Subject": row["subject_id"],
        "FF_T0": ff_T0,
        "FF_T1": ff_T1,
        "FF_T2": ff_T2
    })

# Convert results into a DataFrame
ff_interpolated_df = pd.DataFrame(interpolated_results)

display(ff_interpolated_df)

print("Participants included:", len(ff_interpolated_df))
print("\nMissing interpolated values:")
print(ff_interpolated_df.isna().sum())
```

	Subject	FF_T0	FF_T1	FF_T2
0	1-0019	5.977253	6.744922	6.702831
1	1-0021	3.346368	4.939927	4.368648
2	1-0022	5.476659	6.616676	6.526241

3	1-0024	8.298934	8.052057	8.413974
4	1-0025	9.510767	11.464132	11.592851
5	1-0026	11.947834	11.365333	12.039543
6	1-0028	5.450935	5.161722	5.548338
7	1-0029	3.634287	5.117211	4.717329
8	1-0030	16.470253	16.556287	19.593634
9	1-0031	12.690958	10.718186	8.775492
10	13-0001	5.850324	6.091684	NaN
11	13-0002	4.723446	4.299326	4.118666
12	13-0003	6.489562	6.666532	7.528305
13	13-0004	7.528644	10.666969	11.873496
14	13-0005	9.988280	9.439744	9.778295
15	13-0006	70.878816	72.225970	72.616489
16	13-0007	14.441094	15.137669	18.330662
17	13-0008	73.308772	74.516881	69.891597
18	13-0009	11.046289	10.273535	13.586845
19	13-0010	61.821145	66.902256	69.929863
20	13-0011	7.407480	6.996615	7.279491
21	32-0020	63.715792	67.400795	NaN
22	32-0022	9.459124	7.937283	NaN
23	32-0023	8.661187	9.830487	10.521604
24	32-0024	7.188731	7.228786	7.268230
25	32-0025	46.869114	48.966098	54.689924
26	32-0026	6.069757	6.957668	7.322221
27	32-0027	7.505872	6.065311	7.563943
28	32-0028	37.457323	44.368240	NaN
29	32-0029	53.246534	56.104413	NaN
30	32-0030	6.016090	6.460587	5.024417

```
Participants included: 31

Missing interpolated values:
Subject    0
FF_T0      0
FF_T1      0
FF_T2      5
dtype: int64
```

In [16]:
```python
# Calculate changes in MRI fat fraction

ff_interpolated_df["Change_T1"] = (
    ff_interpolated_df["FF_T1"] - ff_interpolated_df["FF_T0"]
)

ff_interpolated_df["Change_T2"] = (
    ff_interpolated_df["FF_T2"] - ff_interpolated_df["FF_T0"]
)

# Calculate relative percentage changes
# Avoid dividing by zero when baseline fat fraction is zero
baseline = ff_interpolated_df["FF_T0"].replace(0, np.nan)

ff_interpolated_df["Percent_Change_T1"] = (
    ff_interpolated_df["Change_T1"] / baseline * 100
)

ff_interpolated_df["Percent_Change_T2"] = (
    ff_interpolated_df["Change_T2"] / baseline * 100
)

# Summarize changes across participants
summary = pd.DataFrame({
    "Year 1": [
        ff_interpolated_df["Change_T1"].count(),
        ff_interpolated_df["Change_T1"].mean(),
        ff_interpolated_df["Change_T1"].std(),
        ff_interpolated_df["Percent_Change_T1"].mean(),
        ff_interpolated_df["Percent_Change_T1"].std()
    ],
    "Year 2": [
        ff_interpolated_df["Change_T2"].count(),
        ff_interpolated_df["Change_T2"].mean(),
        ff_interpolated_df["Change_T2"].std(),
        ff_interpolated_df["Percent_Change_T2"].mean(),
        ff_interpolated_df["Percent_Change_T2"].std()
    ]
}, index=[
    "Number of Participants",
    "Mean Absolute Change (percentage points)",
```

```
    "SD Absolute Change (percentage points)",
    "Mean Relative Change (%)",
    "SD Relative Change (%)"
])

display(summary.round(2))
```

	Year 1	Year 2
Number of Participants	31.00	26.00
Mean Absolute Change (percentage points)	0.93	1.26
SD Absolute Change (percentage points)	1.91	2.69
Mean Relative Change (%)	6.05	10.30
SD Relative Change (%)	15.99	17.92

Longitudinal MRI Fat Fraction Results

To evaluate changes in MRI fat fraction across participants, right-side fat fraction measurements were linearly interpolated to common comparison times of 11.15 months (Year 1) and 22.13 months (Year 2).

At Year 1, 31 participants had sufficient measurements for interpolation. The mean absolute change in fat fraction was +0.93 percentage points, with a standard deviation of 1.91 percentage points. The mean relative change was +6.05%, with a standard deviation of 15.99%.

At Year 2, 26 participants had sufficient measurements for interpolation. The mean absolute change was +1.26 percentage points, with a standard deviation of 2.69 percentage points. The mean relative change was +10.30%, with a standard deviation of 17.92%.

These findings suggest that MRI fat fraction generally increased over the observation period, consistent with progressive fatty replacement of skeletal muscle in FSHD. However, the relatively large standard deviations indicate substantial variation in progression between participants.

The smaller Year 2 sample size also limits direct comparisons between the two time points. Additional analyses of muscle strength and functional measurements are necessary to determine whether MRI fat fraction provides a more consistent measure of disease progression than other potential clinical-trial endpoints.

In [17]:
```
# Display strength and functional measurement columns
for col in clinical_df.columns:
    if any(term in col.lower() for term in [
        "strength", "dorsi", "tug", "wr_10",
        "ascend", "descend", "severity"
    ]):
        print(col)
```

```
dorsiflexion_qmt_R_baseline
dorsiflexion_qmt_R_y1
dorsiflexion_qmt_R_y2
dorsiflexion_qmt_L_baseline
dorsiflexion_qmt_L_y1
dorsiflexion_qmt_L_y2
tug_time_baseline
tug_time_y1
tug_time_y2
wr_10_baseline
wr_10_y1
wr_10_y2
ascend_4_baseline
ascend_4_y1
ascend_4_y2
descend_4_baseline
descend_4_y1
descend_4_y2
clinical_severity_score_baseline
clinical_severity_score_y1
clinical_severity_score_y2
```

In [18]:
```
# Longitudinal interpolation of strength and functional measurements

measurements = {
    "Dorsiflexion Strength (Right)": "dorsiflexion_qmt_R",
    "Dorsiflexion Strength (Left)": "dorsiflexion_qmt_L",
    "Timed Up and Go": "tug_time",
    "Walking/Running (WR-10)": "wr_10",
    "Stair Ascent": "ascend_4",
    "Stair Descent": "descend_4",
    "Clinical Severity Score": "clinical_severity_score"
}

functional_results = []

for measurement_name, prefix in measurements.items():

    # Strength measurements use a different column naming pattern
    if "dorsiflexion" in prefix:
        columns = [
            f"{prefix}_baseline",
```

```
            f"{prefix}_y1",
            f"{prefix}_y2"
        ]
    else:
        columns = [
            f"{prefix}_baseline",
            f"{prefix}_y1",
            f"{prefix}_y2"
        ]

    for _, row in clinical_df.iterrows():

        times = np.array([
            row["lag_months_baseline"],
            row["lag_months_y1"],
            row["lag_months_y2"]
        ], dtype=float)

        values = np.array([
            row[columns[0]],
            row[columns[1]],
            row[columns[2]]
        ], dtype=float)

        valid = np.isfinite(times) & np.isfinite(values)

        times = times[valid]
        values = values[valid]

        if len(times) < 2:
            continue

        order = np.argsort(times)
        times = times[order]
        values = values[order]

        interpolation_function = interp1d(
            times,
            values,
            kind="linear",
            bounds_error=False,
            fill_value=np.nan
        )

        value_T0 = float(interpolation_function(T0))
        value_T1 = float(interpolation_function(T1))
        value_T2 = float(interpolation_function(T2))

        functional_results.append({
            "Subject": row["subject_id"],
```

```
            "Measurement": measurement_name,
            "T0": value_T0,
            "T1": value_T1,
            "T2": value_T2
        })

functional_df = pd.DataFrame(functional_results)

# Calculate absolute changes
functional_df["Change_T1"] = functional_df["T1"] - functional_df["T0"]
functional_df["Change_T2"] = functional_df["T2"] - functional_df["T0"]

# Calculate relative percentage changes
baseline = functional_df["T0"].replace(0, np.nan)

functional_df["Percent_Change_T1"] = (
    functional_df["Change_T1"] / baseline * 100
)

functional_df["Percent_Change_T2"] = (
    functional_df["Change_T2"] / baseline * 100
)

# Summarize results for each measurement
functional_summary = functional_df.groupby("Measurement").agg(
    N_Year1=("Percent_Change_T1", "count"),
    Mean_Change_Year1=("Percent_Change_T1", "mean"),
    SD_Year1=("Percent_Change_T1", "std"),
    N_Year2=("Percent_Change_T2", "count"),
    Mean_Change_Year2=("Percent_Change_T2", "mean"),
    SD_Year2=("Percent_Change_T2", "std")
)

display(functional_summary.round(2))
```

Measurement	N_Year1	Mean_Change_Year1	SD_Year1	N_Year2	Mean_Change_Year2
Clinical Severity Score	30	2.74	19.12	24	4.18
Dorsiflexion Strength (Left)	26	8.12	31.81	23	9.28
Dorsiflexion Strength (Right)	26	4.94	27.63	23	11.01
Stair Ascent	33	3.80	18.85	27	1.78
Stair Descent	33	2.20	19.47	27	-0.54
Timed Up and Go	33	2.28	13.32	27	-0.81
Walking/Running (WR-10)	33	2.06	10.38	27	3.35

Comparison of Potential Clinical Trial Endpoints

To compare potential clinical trial endpoints, longitudinal changes were calculated for MRI fat fraction, dorsiflexion muscle strength, functional performance, and clinical severity. Measurements were linearly interpolated to common time points of 11.15 months (Year 1) and 22.13 months (Year 2). Mean relative percentage changes and standard deviations were then calculated across participants.

Among the measurements analyzed, the walking/running assessment (WR-10) demonstrated the lowest standard deviation of relative percentage change at both time points. Its standard deviation was 10.38% at Year 1 and 12.11% at Year 2, suggesting relatively consistent changes between participants. However, its mean changes were comparatively small, at +2.06% and +3.35%, respectively.

MRI fat fraction demonstrated larger mean relative changes, increasing by 6.05% at Year 1 and 10.30% at Year 2. Its standard deviations were 15.99% and 17.92%, respectively. These findings suggest that MRI fat fraction may be useful for detecting structural changes associated with FSHD progression, although the variation between participants remains important.

Dorsiflexion strength measurements demonstrated substantially greater variability. Standard deviations ranged from 27.63% to 41.43%, indicating that changes in muscle strength were less consistent across participants when expressed as relative

percentages. Mean strength changes were positive, so these results did not demonstrate a consistent decline in strength over the observation period.

Other functional measurements, including Timed Up and Go and stair ascent and descent, showed relatively small mean changes and varying directions of change. This makes it difficult to identify a consistent pattern of functional decline using these measurements alone.

Overall, the results suggest that MRI fat fraction and WR-10 may be useful exploratory endpoints for future FSHD clinical trials. MRI fat fraction demonstrated larger average relative changes, while WR-10 demonstrated lower variability. Combining structural MRI measurements with functional assessments may provide a more comprehensive evaluation of disease progression than relying on a single measurement.

Conclusions

The goal of this project was to investigate how different measurements can be used to monitor disease progression in facioscapulohumeral muscular dystrophy (FSHD) and identify potential exploratory endpoints for future clinical trials. Our analysis included MRI fat fraction measurements from a single participant's tibialis anterior muscle and longitudinal clinical data from a cohort of 34 participants. Together, these analyses provided insight into changes in muscle composition, muscle strength, and functional performance over time.

MRI Fat Fraction Analysis

Our MRI image analysis demonstrated an increase in fat fraction between two time points. The average fat fraction across eight sampled locations in the left tibialis anterior muscle increased from 42.26% at TP1 to 54.41% at TP2, representing an increase of 12.15 percentage points.

We also used linear interpolation to estimate fat fraction at 45% of the muscle length. At this location, fat fraction increased from 25.40% at TP1 to 46.30% at TP2, an increase of 20.90 percentage points. This difference demonstrates that changes in muscle composition may vary considerably depending on the location within the muscle. Therefore, consistent imaging locations and standardized measurement methods are important when comparing MRI measurements across visits.

Longitudinal Clinical Data Analysis

To evaluate disease progression across participants, we performed linear interpolation of longitudinal clinical measurements to common comparison times of 11.15 months (Year 1) and 22.13 months (Year 2). We then calculated mean relative percentage changes and standard deviations for MRI fat fraction, dorsiflexion strength, functional performance, and clinical severity.

Right-side MRI fat fraction demonstrated a mean relative increase of 6.05% at Year 1 and 10.30% at Year 2, with standard deviations of 15.99% and 17.92%, respectively. These results suggest that MRI fat fraction may be useful for detecting structural changes associated with disease progression. However, the standard deviations indicate that the magnitude of these changes varied between participants.

Among the functional measurements analyzed, the walking/running assessment (WR-10) demonstrated the lowest standard deviation of relative percentage change at both time points. Its standard deviation was 10.38% at Year 1 and 12.11% at Year 2, with mean relative changes of 2.06% and 3.35%, respectively. This suggests that WR-10 exhibited relatively consistent changes across participants, although the average changes were smaller than those observed for MRI fat fraction.

Dorsiflexion strength measurements demonstrated substantially greater variability, with standard deviations ranging from 27.63% to 41.43%. Additionally, the average strength measurements increased rather than decreased over time, which does not reflect the expected pattern of progressive muscle weakness. These findings may reflect differences between participants, measurement variability, or other factors that cannot be determined from the available data alone.

Other functional measurements, including Timed Up and Go and stair ascent and descent, demonstrated relatively small average changes and inconsistent directions of change. These findings suggest that functional measurements alone may not consistently capture disease progression over the observation period.

Recommendations for Future Clinical Trials

Based on our analysis, MRI fat fraction and WR-10 appear to be promising exploratory endpoints for future FSHD clinical trials, although for different reasons.

MRI fat fraction demonstrated relatively large average changes over time and provides a quantitative measurement of structural changes within skeletal muscle. This makes it potentially useful for detecting disease-related changes that may occur before noticeable differences in physical performance.

WR-10 demonstrated the lowest variability in relative percentage changes among the measurements analyzed. Although its average changes were smaller, functional assessments provide information about physical abilities that are directly relevant to patients' daily lives.

Therefore, combining MRI fat fraction with functional assessments such as WR-10 may provide a more comprehensive evaluation of FSHD progression than relying on either measurement alone. MRI can provide information about structural muscle changes, while functional measurements can help determine whether those changes are associated with meaningful differences in physical performance.

Additional Evidence Needed

Before these measurements could be used as reliable clinical trial endpoints, additional research would be necessary. Larger longitudinal studies would help determine whether the observed patterns remain consistent across a broader FSHD population.

It would also be important to evaluate the relationship between MRI fat fraction changes and functional outcomes, including muscle strength, walking performance, and clinical severity. Establishing these relationships could help determine whether changes in MRI measurements correspond to clinically meaningful changes in patients' abilities.

Additionally, studies involving treated and untreated participants would be necessary to determine whether these measurements can distinguish treatment effects from natural disease progression. Measurement reproducibility, differences between muscle groups, and the effects of missing data should also be investigated.

Ethical Considerations

Selecting clinical trial endpoints involves ethical considerations because the chosen measurements influence how treatment effectiveness is evaluated.

MRI provides detailed information about muscle composition, but it can be expensive and may not be equally accessible to all patients. Functional assessments may be more accessible and directly relevant to daily activities, but their results can be influenced by fatigue, motivation, and differences in physical ability.

Relying exclusively on MRI measurements could overlook improvements in strength or mobility that are meaningful to patients. Conversely, relying only on functional assessments could fail to detect early structural changes associated with disease progression.

For these reasons, clinical trials should consider multiple complementary measurements and prioritize outcomes that reflect both biological changes and improvements in patients' quality of life.

Overall Conclusion

Our findings suggest that MRI fat fraction is a promising measurement for monitoring structural changes associated with FSHD, while WR-10 demonstrated relatively consistent changes across participants. However, neither measurement alone provides a complete assessment of disease progression.

A combination of MRI, muscle strength, and functional assessments may provide a more comprehensive approach to evaluating FSHD progression and potential treatment effectiveness. Further research with larger participant groups, standardized measurement methods, and clinically meaningful outcomes is needed before determining which endpoints are most appropriate for future clinical trials.

Limitations and Future Work

Although our analysis provided insight into potential measurements of FSHD progression, several limitations should be considered when interpreting the results.

Sample Size and Missing Data

One limitation of this study was the relatively small dataset of 34 participants. Although the dataset included multiple measurements of muscle composition, strength, and functional performance, not every participant had complete measurements at all three time points.

For example, only 32 participants had recorded Year 1 visit times, and 27 had recorded Year 2 visit times. In our longitudinal MRI fat fraction analysis, 31 participants had sufficient data for interpolation at Year 1, while only 26 had sufficient data at Year 2.

These missing measurements reduced the number of participants included in our calculations and may have influenced the observed averages and standard deviations. Additionally, differences in the participants included at each time point limit direct comparisons between Year 1 and Year 2 results.

Future studies should include larger participant groups and prioritize consistent follow-up measurements to reduce missing data and improve the reliability of longitudinal

comparisons.

Limitations of Linear Interpolation

Our longitudinal analysis used linear interpolation to estimate measurements at common comparison times of 11.15 months and 22.13 months. This method allowed us to compare participants whose visits occurred at slightly different times.

However, linear interpolation assumes that measurements change at a constant rate between observed time points. FSHD progression may not occur linearly, and individual participants may experience periods of faster or slower progression.

Additionally, interpolation cannot reliably estimate measurements outside the observed time range without extrapolation. We avoided extrapolating beyond the available measurements, which resulted in some missing interpolated values.

Future analyses could compare linear interpolation with alternative approaches, such as quadratic interpolation or longitudinal statistical models, to determine which methods best represent changes over time. These approaches would require sufficient observations to avoid introducing additional uncertainty.

MRI Analysis Limitations

Our detailed MRI image analysis focused on the left tibialis anterior muscle of a single participant at two time points. Although we observed an increase in average fat fraction from 42.26% to 54.41%, this result cannot be generalized to the entire FSHD population.

We also observed that fat fraction varied across different locations within the muscle. Our spatial interpolation at 45% muscle length produced a larger increase than the average across all eight sampled locations, demonstrating that measurement location can influence the estimated degree of muscle change.

Additionally, our overall MRI averages were calculated by giving equal weight to each sampled muscle location rather than weighting measurements by muscle area or volume. This means the averages do not necessarily represent the true fat fraction of the entire muscle.

Future studies should examine MRI data from multiple participants and muscle groups. More comprehensive three-dimensional measurements, standardized imaging locations, and volume-weighted calculations could provide a more representative assessment of muscle composition.

Variability Between Participants

Our longitudinal analysis demonstrated considerable variability in disease-related measurements.

For example, MRI fat fraction showed mean relative increases of 6.05% at Year 1 and 10.30% at Year 2, but the corresponding standard deviations were 15.99% and 17.92%.

Dorsiflexion strength measurements demonstrated even greater variability, with standard deviations ranging from 27.63% to 41.43%.

These results indicate that participants did not experience identical changes over time. Differences in disease severity, affected muscle groups, individual progression rates, and measurement variability may contribute to these differences.

Future research could investigate whether factors such as age, baseline disease severity, and initial MRI fat fraction are associated with different rates of progression.

Limitations of Comparing Clinical Trial Endpoints

Our comparison of potential clinical trial endpoints was based primarily on mean relative percentage changes and their standard deviations.

Although WR-10 demonstrated the lowest standard deviation at both time points, lower variability does not necessarily indicate greater sensitivity to disease progression or treatment effects.

Relative percentage changes can also become unstable when baseline measurements are close to zero. Additionally, the measurements included in our analysis represent different aspects of FSHD and may not be directly comparable based on percentage change alone.

Our analysis did not establish whether changes in MRI fat fraction were associated with changes in muscle strength or functional performance. Therefore, we cannot determine whether the observed structural changes corresponded to clinically meaningful differences in participants' physical abilities.

Future analyses could examine correlations between MRI fat fraction and functional measurements, evaluate measurement reproducibility, and compare the ability of different endpoints to detect meaningful changes over time.

Future Research and Clinical Applications

Future studies should investigate whether combining structural MRI measurements with functional assessments provides a more reliable evaluation of FSHD progression than using individual measurements.

MRI fat fraction may help identify changes in muscle composition, while strength and functional assessments can provide information about physical performance and limitations in daily activities.

It would also be valuable to investigate whether baseline molecular measurements, such as DUX4 gene expression signatures, are associated with subsequent changes in MRI fat fraction or functional performance. Because DUX4 signatures were only available at baseline in our dataset, we could not evaluate their longitudinal changes.

Finally, studies comparing treated and untreated participants would be necessary to determine whether these measurements can distinguish treatment effects from natural disease progression.

Overall, addressing these limitations through larger datasets, standardized measurement methods, and more comprehensive longitudinal analyses could improve the identification of reliable and clinically meaningful endpoints for future FSHD clinical trials.

AI Usage Statement

We used ChatGPT (OpenAI) to assist with developing Python code, interpreting results, organizing project sections, and drafting written explanations.

ChatGPT assisted with loading MRI images, applying segmentation masks, calculating fat fractions, generating visualizations, and implementing spatial and longitudinal interpolation. It also helped develop the comparison of clinical trial endpoints using mean percentage changes and standard deviations.

We executed the code in Jupyter Notebook using Visual Studio Code and reviewed the resulting calculations, tables, and figures. We also verified that all 16 segmentation masks contained binary values.

ChatGPT assisted with drafting the Conclusions and Limitations and Future Work sections based on the calculated results. It also helped with source research, citation formatting, and organizing the written sections.

The data used in this project were provided through BME 2315. The reported

calculations and conclusions are based on the supplied data and the referenced course materials.

GitHub Repository Link:

https://github.com/tkt6gs/BME2315_Module_2.git

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Notes from Your Team

Team Responsibilities

Imtiage Ahamed:

- Researched FSHD and its biological mechanisms.
- Completed the Project Goal and Disease Background sections.
- Gathered supporting references.

Liam Sunshine:

- Documented the clinical dataset and MRI analysis methodology.
- Processed MRI images and calculated muscle fat fractions.
- Performed spatial and longitudinal interpolation.
- Compared MRI, muscle strength, and functional measurements.
- Completed the Conclusions and Limitations and Future Work sections.

Project Activity Log

Initial Project Development

- Reviewed the project requirements and clinical dataset.
- Divided responsibilities between team members.
- Researched FSHD and potential clinical trial endpoints.

MRI Data Analysis

- Loaded MRI water, fat, and segmentation mask files.
- Calculated fat fractions for eight muscle locations at two time points.
- Visualized fat fraction distributions and changes between visits.
- Used linear interpolation to estimate fat fraction at 45% muscle length.

Longitudinal Analysis

- Examined the clinical dataset containing 34 participants.
- Identified missing measurements and differences in visit times.
- Interpolated measurements to common time points of 11.15 and 22.13 months.
- Calculated mean percentage changes and standard deviations.
- Compared MRI fat fraction, muscle strength, and functional assessments.

Final Project Preparation

- Interpreted the results and identified promising exploratory endpoints.
- Documented limitations and potential directions for future research.
- Combined the project sections into the final Jupyter notebook.

QUESTIONS FOR YOUR TA:

We have no questions for our TA